

Monitoring Bone Fracture Healing Using Guided Acoustic Waves Towards a Wearable Ultrasound System

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Background, Motivation and Objective

Bone fractures represent an increasing medical and social challenge, particularly due to falls and osteoporosis in the elderly population. Among these injuries, tibial fractures are of particular clinical importance because of their frequency and the often prolonged healing process. For effective clinical management, it is essential to determine whether biological healing progresses properly and whether sufficient stiffness and stability of the bone have been restored, for example when deciding the appropriate time to remove fixation devices.

Currently, fracture healing is mainly monitored using functional assessments and X-ray imaging. However, radiographic imaging involves exposure to ionizing radiation, is relatively costly, and does not provide direct information about the mechanical stiffness of the healing bone. Therefore, alternative methods capable of assessing the mechanical state of the fracture region in a non-invasive and repeatable manner are highly desirable.

In this work, we investigate an ultrasonic approach based on guided acoustic waves to monitor elasticity changes in the fracture region during the healing process.

Statement of Contribution/Methods

The proposed method analyzes the propagation characteristics of guided acoustic Lamb waves traveling through the fractured bone region. As healing progresses, the mechanical properties of the fracture zone change, which affects the propagation behavior of these waves.

To study this effect, bone phantoms replicating different anatomical geometries and fracture types were developed. Initial experiments were performed using a single-element ultrasonic transmitter–receiver configuration. The received signals were evaluated with respect to wave propagation characteristics and compared with theoretical predictions for guided wave behavior.

Results/Discussion

The initial measurements show that the propagation characteristics of guided acoustic waves are sensitive to changes in the mechanical properties of the fracture region. The experimental observations demonstrate good agreement with theoretical expectations.

These results indicate that the proposed method may enable monitoring of fracture healing by tracking elasticity changes in the bone and potentially allow prediction of the remaining healing time. Future work will focus on increasing the realism of the phantoms by including soft-tissue layers and on advancing both hardware and signal processing toward the development of a wearable ultrasound device for continuous fracture monitoring.